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# ON THE COMPUTATION OF ANGULAR DISTRIBUTIONS OF RADIATION IN PLANETARY ATMOSPHERES

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**Abstract**—The discrete ordinate approximation to the radiative transfer equation is used to derive simple analytic expressions for the intensity in a vertically inhomogeneous plane parallel atmosphere. This approach allows one to compute the intensity at arbitrary depths and angles. The merit and soundness of these expressions, which are interpolatory in nature, are discussed and it is conjectured that they are superior to any other standard interpolation scheme as far as accuracy is concerned. The computational time also compares favorably with that of standard interpolation schemes such as cubic splines.

## 1. INTRODUCTION

In a recent paper dealing with intensity computations by the spherical harmonic method, Karp[1] discusses various interpolation schemes for computing the specific intensity at angles other than those corresponding to the quadrature angles (in the discrete ordinate sense). Although the spherical harmonic method specifies the intensity at any angle, Karp[1] notes that the intensity computed by this method exhibits an oscillatory behavior, and he shows (by comparing with the discrete ordinate method) that in the azimuth-independent case it is expected to be accurate only at the quadrature points. Thus, it would be desirable to employ an interpolation scheme based on information available at the quadrature points to compute the intensity at angles other than the quadrature angles. After comparing the integration of the source function approach to intensity computations with cubic spline interpolations, Karp[1] concludes that the former approach is by far the most accurate.

In two recent papers,[2,3] it was shown that the discrete ordinate method, when properly executed, is indeed a very versatile, reliable and efficient technique for computing intensities and fluxes in planetary atmospheres. Perhaps not surprisingly, and in spite of being developed independently, the numerical techniques used to find the eigensolutions in the discrete ordinate method,[2-4] are, in fact, very similar to those adopted by Karp *et al.*[5] for the spherical harmonic method.

The primary purpose of the present paper is to provide simple analytic expressions for the intensity in a vertically inhomogeneous (multi-layered) plane parallel atmosphere within the framework of the discrete ordinate method. While the present approach is also based on the integration of the source function technique,[2-4] it avoids the matrix inversions invoked by Karp[1] in the spherical harmonic method by providing analytic expressions for the intensity. From these expressions, the intensity can be computed at arbitrary depths; the method of Karp only gives intensities at layer interfaces. A discussion of the merit of the present approach is also provided.

## 2. ANALYTIC EXPRESSIONS FOR THE INTENSITY IN AN INHOMOGENEOUS ATMOSPHERE

We consider an inhomogeneous atmosphere that can be adequately described by a series of $N$ adjacent, homogeneous layers in which the scattering and absorption properties are allowed to vary from layer to layer. Let $I_i^m(\tau, \mu)$ denote the $m$th Fourier component of the intensity[3] at level $\tau$ such that $\tau_{i-1} \leq \tau \leq \tau_i$. The general discrete ordinate solution, for each Fourier component of the diffuse intensity for an arbitrary layer denoted by $p$ due to a parallel beam of radiation incident in direction $\theta_0 = \cos^{-1} \mu_0$, can then be written in the compact form[3]

$$I_p^m(\tau, \mu_s) = \sum_{j=-n}^{n} G_{jp}^m(\mu_s)\, e^{-k_{jp}^m \tau}, \tag{1}$$

where $G_{jp}^m(\mu_s) = L_{jp}^m g_{jp}^m(\mu_s)$ for $j \neq 0$. The $k_{jp}^m$ and the $g_{jp}^m(\mu_s)$ are the eigenvalues and eigenvectors, the $\mu_s$ are the quadrature angles and the $L_{jp}^m$ are constants of integration.[2,3] The sum in Eq. (1) runs from $-n$ to $+n$ including $j = 0$. Thus, we define

$$G_{0p}^m(\mu_s) \equiv Z_{0p}^m(\mu_s), \tag{2a}$$

$$k_{0p}^m \equiv 1/\mu_0, \tag{2b}$$

where $Z_{0p}^m(\mu_s)\, e^{-\tau/\mu_0}$ is the particular solution.[3] To facilitate the subsequent discussion, it is also convenient to define the function

$$S_p^m(\tau, \mu) \equiv \sum_{j=-n}^{n} G_{jp}^m(\mu)\, e^{-k_{jp}^m \tau}. \tag{3}$$

The meaning of this function will be made clear in Section 4. The intensity given by Eq. (1) must satisfy boundary conditions at the top ($\tau = 0$) and the bottom ($\tau = \tau_N$) of the atmosphere as well as continuity conditions at layer interfaces, as follows:[6,7]

$$\left.\begin{aligned} I_1^m(0, -\mu_s) &= 0,\ s = 1, \ldots, n, \\ I_p^m(\tau_p, \mu_s) &= I_{p+1}^m(\tau_p, \mu_s),\ s = \pm 1, \ldots \pm n;\ p = 1, \ldots, N-1. \\ I_N^m(\tau_N, +\mu_s) &= 0,\ s = 1, \ldots, n. \end{aligned}\right\} \tag{4a}$$

In terms of $S_p^m(\tau, \mu)$, defined by Eq. (3), these boundary and continuity conditions may also be expressed as

$$\left.\begin{aligned} S_1^m(0, -\mu_s) &= 0,\ s = 1, \ldots, n. \\ S_p^m(\tau_p, \mu_s) &= S_{p+1}^m(\tau_p, \mu_s),\ s = \pm 1, \ldots, \pm n;\ p = 1, \ldots, N-1, \\ S_N^m(\tau_N, +\mu_s) &= 0,\ s = 1, \ldots, n. \end{aligned}\right\} \tag{4b}$$

The formal solutions for the intensity are given by[2]

$$\begin{aligned} I_i^m(\tau, -\mu) &= \int_0^{\tau} J^m(t, -\mu)\, e^{-(\tau - t)/\mu}\, dt/\mu \\ &= \sum_{p=1}^{i-1} \int_{\tau_{p-1}}^{\tau_p} J_p^m(t, -\mu)\, e^{-(\tau - t)/\mu}\, dt/\mu + \int_{\tau_{i-1}}^{\tau} J_i^m(t, -\mu)\, e^{-(\tau - t)/\mu}\, dt/\mu \end{aligned} \tag{5}$$

$$\begin{aligned} I_i^m(\tau, +\mu) &= \int_{\tau}^{\tau_N} J^m(t, \mu)\, e^{-(t - \tau)/\mu}\, dt/\mu \\ &= \int_{\tau}^{\tau_i} J_i^m(t, \mu)\, e^{-(t - \tau)/\mu}\, dt/\mu + \sum_{p=i+1}^{N} \int_{\tau_{p-1}}^{\tau_p} J_p^m(t, \mu)\, e^{-(t - \tau)/\mu}\, dt/\mu. \end{aligned} \tag{6}$$

Since the source function for each layer can readily be expressed as[2-4]

$$J_p^m(t, \mu) = \sum_{j=-n}^{n} (1 + k_{jp}^m \mu) G_{jp}^m(\mu)\, e^{-k_{jp}^m t}, \tag{7}$$

Eqs. (5) and (6) lead to the following simple expressions for the intensity:

$$I_i^m(\tau, -\mu) = \sum_{p=1}^{i} \left[ \sum_{j=-n}^{n} G_{jp}^m(-\mu) \{ e^{-[k_{jp}^m \tau_p + (\tau - \tau_p)/\mu]} - e^{-[k_{jp}^m \tau_{p-1} + (\tau - \tau_{p-1})/\mu]} \} \right] \tag{8}$$

with $\tau_p = \tau$ for $p = i$,

$$I_i^m(\tau, +\mu) = \sum_{p=i}^{N} \left[ \sum_{j=-n}^{n} G_{jp}^m(\mu) \{ e^{-[k_{jp}^m \tau_{p-1} - (\tau - \tau_{p-1})/\mu]} - e^{-[k_{jp}^m \tau_p - (\tau - \tau_p)/\mu]} \} \right] \tag{9}$$

with $\tau_{p-1} = \tau$ for $p = i$.

## 3. INTERPOLATION

It is evident from Eqs. (8) and (9) that the merit of these expressions depends crucially on the ability to compute efficiently (for each layer $p$) the eigenfunctions, $G^m_{jp}(\mu)$ or, in view of Eq. (2), the eigenvectors $g^m_{jp}(\mu)$'s and the particular solution vector $Z^m_{0p}(\mu)$. If this can be accomplished, then Eqs. (8) and (9) provide a convenient and efficient means of computing the intensities for arbitrary angles, at any desired level, $\tau$.

Since the $G^m_{jp}(\mu)$ are known at the quadrature points ($\mu = \mu_s$, $s = \pm 1, \ldots, \pm n$), this information can be used as a basis for interpolation using any standard interpolation scheme. However, very simple analytic expressions for the $g^m_{jp}(\mu)$ and $Z^m_{0p}(\mu)$, involving the values at the quadrature points, can be derived from the basic transfer equation to which we are seeking a solution.[2,3] This fact suggests that these expressions may be superior to any other standard interpolation scheme. In Ref. 3, we demonstrated that this approach provides an efficient and reliable means of computing the complete azimuthal dependence of the intensity.

To illustrate the problems one might encounter in interpolation (using standard techniques) we show in Fig. 1 for a 16-stream computation (i.e. $n = 8$) the eigenvector corresponding to the smallest eigenvalue illustrating the typical behavior of some of the eigenvectors for the Haze $L$ phase function of Deirmendjian[8] (as reported by Lenoble[9]) with single scattering albedo $\omega_0 = 0.9$. The broken curve corresponds to cubic spline interpolation and the solid curve to our analytic expressions.[2,3] The data points to be interpolated are indicated by the dots. As one might expect, the cubic spline performs poorly on both sides of the dip, whereas our analytic expressions yield quite adequate results.

The consequence of using these different interpolation schemes is perhaps best illustrated by giving results for a homogeneous atmosphere since the performance of these schemes is unrelated to the inhomogeneity. Thus, Fig. 2 shows the resulting diffuse intensities at the top and bottom of a homogeneous slab of optical thickness $\tau^* = 1$ for a parallel beam of flux $\mu_0$ (i.e. unit incident intensity) incident perpendicular to the slab ($\mu_0 = 1$). Expressions (6) and (7) of Ref. 3 were used to generate these curves. The solid lines are based on our analytic expressions [Eqs. (8) and (9) of Ref. 3] to compute the eigenvectors which yield an accuracy of a few percent for 16 streams.[3] The results obtained using cubic spline interpolation of the eigenvectors are shown by the broken lines. This approach yields inaccurate values for the reflected intensity for $0 < \mu < 0.4$. Since these intensity curves are quite smooth, a straightforward interpolation based on the intensities at the quadrature points (shown by dots) should give adequate results as demonstrated by the dotted lines.

However, the intensities are not always as smooth as those shown in Fig. 2.[1,3] Thus, repeating the calculation changing only the angle of incidence so that $\mu_0 = 0.5$, one gets the

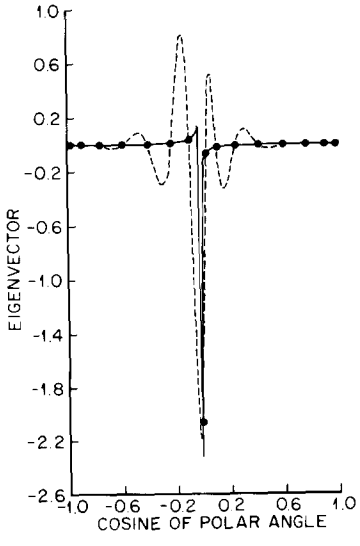



Fig. 1. Interpolated eigenvector; — analytic expressions; --- cubic spline interpolation, ● values at quadrature points.

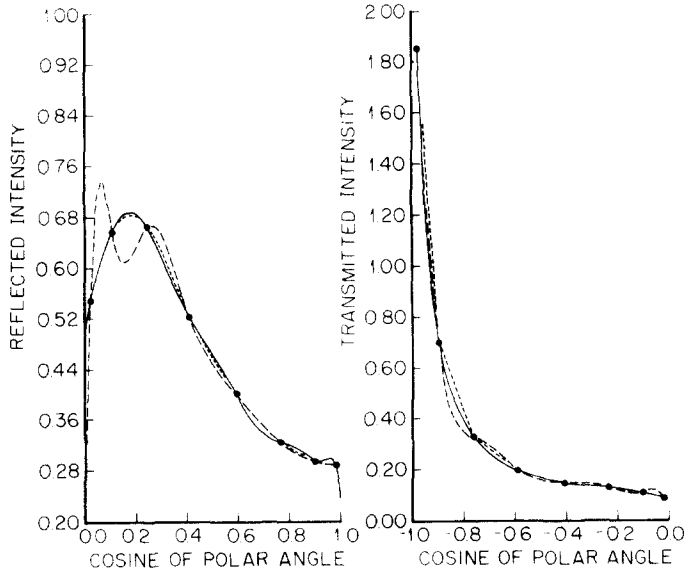



Fig. 2. Reflected and transmitted intensities for normal incidence ($\mu_0 = 1.$) of unit intensity on a slab of optical thickness $\tau^* = 1$ that scatters radiation according to the Haze $L$ phase function with single scattering albedo $\omega_0 = 0.9$. — analytic expressions, --- cubic spline interpolation of eigenvectors, · · · cubic spline interpolations of intensities at quadrature points, ● intensities at quadrature points.

curves shown in Fig. 3 for the azimuthally-averaged intensity at optical depth $\tau = 0.05$. The cubic spline interpolation leads to erroneous results for $-0.6 < \mu < -0.1$.

## 4. DISCUSSION

In order to demonstrate the basic soundness of the intensity expressions, it is convenient to rewrite Eqs. (8) and (9) as follows [using Eq. (3)]:

$$I_i^m(\tau, -\mu) = S_i^m(\tau, -\mu) - S_1^m(0, -\mu)\, e^{-\tau/\mu} + \sum_{p=1}^{i-1} \{S_p^m(\tau_p, -\mu) - S_{p+1}^m(\tau_p, -\mu)\} e^{-(\tau-\tau_p)/\mu}, \tag{10}$$

$$I_i^m(\tau, +\mu) = S_i^m(\tau, +\mu) - S_N^m(\tau_N, +\mu)\, e^{-(\tau_N-\tau)/\mu} + \sum_{p=i+1}^{N} \{S_p^m(\tau_{p-1}, +\mu) - S_{p-1}^m(\tau_{p-1}, +\mu)\}\, e^{-(\tau_N-\tau)/\mu}. \tag{11}$$

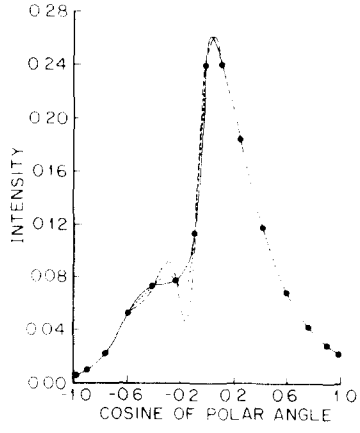



Fig. 3. Azimuthally-averaged intensity at $\tau = 0.05$ for the same situation as in Fig. 2 except that $\mu_0 = 0.5$. See the legend of Fig. 2 for explanation of symbol.

We immediately observe that [in view of Eq. (4b)] Eqs. (10) and (11) coincide with Eq. (1) at the quadrature points, i.e., for $\mu = \mu_s$, Eqs. (10) and (11) reduce to

$$I_i^m(\tau, \mu) = S_i^m(\tau, \mu_s) = \sum_{j=-n}^{n} G_{ji}^m(\mu_s)\, e^{-k_j^m \tau}. \tag{12}$$

This observation clearly indicates the interpolatory nature of Eqs. (8) and (9). Moreover, if we required Eq. (4b) to be satisfied for all $\mu$ (which is generally not the case), then Eqs. (10) and (11) would reduce to Eq. (12) for all $\mu$. Even though Eq. (4b) is not satisfied for all $\mu$, the intensities given by Eqs. (8) and (9), satisfy the boundary and continuity conditions given by Eq. (4a) for all $\mu$. In fact, Kourganoff[10] suggested using the integration of the source function method for exactly this reason.

The role of the function $S_p^m(\tau_p, \mu)$ (evaluated at the boundaries and layer interfaces) in "correcting" the simpler expression (12) (for $\mu \neq \mu_s$) is quite clear from Eqs. (10) and (11). This observation is particularly transparent for a single homogeneous layer (SHL), in which case the sums in Eqs. (10) and (11) disappear. It is also apparent in this case (SHL) that for the downward [upward] intensity this correction, $S_1^m(0, -\mu)[S_N^m(\tau_N, \mu)]$, plays the role of ensuring that Eq. (10) [Eq. (11)] satisfies the boundary conditions for arbitrary values of $\mu$, while the exponential factor $e^{-\tau/\mu}[e^{-(\tau_N-\tau)/\mu}]$, makes a similar correction, appropriately attenuated, at all depths. The correction is more important for large than for small $\mu$-values, while it becomes unnecessary in the limit as $\mu$ approaches zero [in fact, for $\mu = 0$, we have $I_p^m(\tau, 0) = S_p^m(\tau, 0) = J_p^m(\tau, 0)$]. Analogously, for a multi-layered atmosphere, the difference $(S_p^m(\tau_p, -\mu) - S_{p+1}^m(\tau_p, -\mu))[(S_p^m(\tau_{p-1}, +\mu) - S_{p-1}^m(\tau_{p-1}, +\mu))]$ plays a similar role in making Eq. (10) [Eq. (11)] satisfy the continuity conditions for arbitrary values of $\mu$.

As mentioned in the introduction, Karp's strategy for spline interpolation is based on the observation that the azimuth-independent intensity (i.e., the zero-order Fourier component) computed by the spherical harmonic method is accurate at the quadrature points. Whether or not this strategy can be extended to the higher-order Fourier components ($m \neq 0$) is not clear, since the equivalence between the spherical harmonic and the discrete ordinate method has (to our knowledge) only been proven for the azimuth-independent case. Hence, if one desires to use spline interpolations also for higher-order intensity components, one first needs to establish at which angles the spherical harmonic method provides accurate results. Such problems do not occur in the discrete ordinate method, since the quadrature points provide a unique basis for interpolation for all Fourier components of the intensity (regardless of interpolation technique).

It is apparent from Eqs. (8) and (9) that an interpolation scheme which interpolates the eigenvectors, is perhaps best suited as a general purpose interpolation scheme since it can provide intensities at any desired angle and depth. In this respect, we have demonstrated that our analytic expressions are vastly superior to standard cubic spline interpolation as far as accuracy is concerned. The computing time which is almost identical for the two methods, is directly proportional to the number of angles and depths for which the intensity is required.

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